

1. (a) $\Omega \neq \emptyset$, $\mathcal{C} \subset \mathcal{P}(\Omega)$ is semi-algebra if
 - (i) $A, B \in \mathcal{C} \Rightarrow A \cap B \in \mathcal{C}$ (π -system);
 - (ii) $\forall A \in \mathcal{C} \Rightarrow A^C = \bigcup_{i=1}^k B_i$, $\{B_i\}_{i=1}^k \subset \mathcal{C}$ are disjoint;
- (b) $\Omega \neq \emptyset$, $\mathcal{F} \subset \mathcal{P}(\Omega)$ is an algebra if
 - (i) $A, B \in \mathcal{F} \Rightarrow A \cup B \in \mathcal{F}$;
 - (ii) $A \in \mathcal{F} \Rightarrow A^C \in \mathcal{F}$;
 - (iii) $\Omega \in \mathcal{F}$.
- (c) $\Omega \neq \emptyset$, $\mathcal{F} \subset \mathcal{P}(\Omega)$ is an σ -algebra if
 - (i) $\{A_i\} \subset \mathcal{F} \Rightarrow \bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$;
 - (ii) $A \in \mathcal{F} \Rightarrow A^C \in \mathcal{F}$;
 - (iii) $\Omega \in \mathcal{F}$.
- (d) $(\Omega, \mathcal{F})$ is called a **measurable space** if $\Omega \neq \emptyset$ and $\mathcal{F}$ is a σ -algebra.
- (e) A set function μ defined on an algebra/semi-algebra/ σ -algebra $\mathcal{F}$ ($\Omega \neq \emptyset$) is called a measure if
 - (i) $\mu : \mathcal{F} \rightarrow [0, \infty]$;
 - (ii) $\mu(\emptyset) = 0$;
 - (iii) $\forall$ disjoint collection of sets $\{A_i\}_i \subset \mathcal{F}$, $\mu(\bigcup_i A_i) = \sum_{i=1}^{\infty} \mu(A_i)$.
- (f) A measure μ is called **finite** if $\mu(\Omega) < \infty$. A finite measure μ with $\mu(\Omega) = 1$ is called a **probability measure**.
- (g) A measure μ is σ -finite if $\exists \{A_i, i = 1, 2, \dots\} \subset \mathcal{F}$, s.t. $\bigcup_i A_i = \Omega$ and $\mu(A_i) < \infty, \forall i$.
- (h) If μ is a **measure on a measurable space** $(\Omega, \mathcal{F})$, then $(\Omega, \mathcal{F}, \mu)$ is called a **measure space**. In addition, if μ is a probability measure, then $(\Omega, \mathcal{F}, \mu)$ is a **probability space**.
- (i) If $(\Omega_1, \mathcal{F}_1)$ and $(\Omega_2, \mathcal{F}_2)$ are two measurable spaces and if $X : \Omega_1 \rightarrow \Omega_2$ satisfies $X^{-1}(B) \in \mathcal{F}_1, \forall B \in \mathcal{F}_2$, then X is called a measurable transformation.
- (j) If $(\Omega_1, \mathcal{F}_1)$ has a probability measure defined on it, and if $(\Omega_2, \mathcal{F}_2) = (\mathbb{R}, \mathcal{B}(\mathbb{R}))$, then the measurable transformation X is called a random variable.

2. Taking $F = \Omega \in \mathcal{F}$, we get $\Omega_0 = \Omega \cap \Omega_0 \in \mathcal{F}_0$.

If $A = F \cap \Omega_0 \in \mathcal{F}_0$ for some $F \in \mathcal{F}$, then $\Omega_0 \setminus A = F^c \cap \Omega_0 \in \mathcal{F}_0$, since $F^c \in \mathcal{F}$ (note that $\Omega_0 \setminus A$ is the complement of A in Ω_0).

If $\{A_i\} \subset \mathcal{F}_0$, then each $A_i = F_i \cap \Omega_0$, for some $\{F_i\} \subset \mathcal{F}$. In this case, $\bigcup_{i \geq 1} F_i \in \mathcal{F}$ and hence $\bigcup_{i \geq 1} A_i = (\bigcup_{i \geq 1} F_i) \cap \Omega_0 \in \mathcal{F}_0$. This proves that $\mathcal{F}_0$ is a σ -algebra of subsets of Ω_0 .

3. (a) $f = 1_A$, where A is a non-Borel measurable set.
- (b) In the same example, take $g = -f$. Then $h = f + g = 0$ is Borel measurable (constant functions are always measurable).
- (c) Note that for $B \in \mathcal{B}(\mathbb{R})$,

$$T^{-1}(B) = \begin{cases} \emptyset; & \text{if } 5 \notin B \text{ and } 0 \notin B \\ (-\infty, 0]; & \text{if } 5 \in B \text{ and } 0 \notin B \\ (0, \infty); & \text{if } 5 \notin B \text{ and } 0 \in B \\ \mathbb{R}; & \text{if } 5, 0 \in B \end{cases}$$

Since $(-\infty, 0], (0, \infty) \notin \mathcal{F}_1$, then T is not $\langle \mathcal{F}_1, \mathcal{F}_2 \rangle$ measurable.

4. (a) $\mu^*(\emptyset) = 0$, $\mu^*(A) \leq \mu^*(B)$ if $A \subset B$ (monotone) and for any countable collection of subsets of Ω , $\{A_i\}$, the following holds $\mu^*(\bigcup_i A_i) \leq \sum_i \mu^*(A_i)$ (ctble. subadditivity).
- (b) A set $A \in \mathcal{P}(\Omega)$ is called **μ^* -measurable** if: $\mu^*(A) = \mu^*(A \cap E) + \mu^*(A \cap E^c)$ for any $E \in \mathcal{P}(\Omega)$.
- (c) Note that if μ^* is finitely-additive, then $\mu^*(B_1 \cup B_2) = \mu^*(B_1) + \mu^*(B_2)$ for any disjoint B_1, B_2 . Now take any $A \in \mathcal{P}(\Omega)$ and any $E \in \mathcal{P}(\Omega)$. Note that $B_1 = A \cap E$ and $B_2 = A \cap E^c$ are disjoint and hence A is μ^* -measurable. That is, any $A \in \mathcal{P}(\Omega)$ is μ^* -measurable, or in other words, $\mathcal{M}_{\mu^*}$ = set of all μ^* -measurable sets is all of the power set $\mathcal{P}(\Omega)$. But by the theorem in the book (**Theorem 1.3.2**), μ^* is a measure on $\mathcal{M}_{\mu^*}$ and $\mathcal{M}_{\mu^*} = \mathcal{P}(\Omega)$ in this case. So, μ^* is a measure on $\mathcal{P}(\Omega)$.